

Cryptography and Network Security

Integrated Situation — Technical Report

Josue Baraka Rudahalingoma
ULK Polytechnic Institute

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1 Introduction

This report documents the security assessment and technical work carried out for the ETTCS801 Integrated Situation. The scenario involves a polytechnic institute storing student records on a central server and transferring files between two campuses, with weak staff passwords, outdated software, unencrypted transfers, and guest network access to the records server identified as issues in a security review.

2 Risk Assessment

2.1 Assets, Vulnerabilities and Consequences

Asset	Vulnerability	Consequence
Records server	Guest network access	Unauthorised access/tampering
Staff credentials	Weak passwords	Account compromise
File transfer channel	Unencrypted transfer	Interception in transit

Table 1: Assets, vulnerabilities and consequences

2.2 Risk Ranking

2.3 Recommended Controls

3 Encryption and Integrity Toolkit

The `crypto_toolkit.py` programme was developed to encrypt and decrypt student records and to verify their integrity.

3.1 Design

- **Encryption:** symmetric encryption using `Fernet` (AES-128 in CBC mode with HMAC authentication) from the Python `cryptography` library.
- **Integrity:** SHA-256 hashing via `hashlib`, with the hash saved alongside the file and re-checked on demand.
- **Robustness:** all file operations are wrapped in `try/except` blocks so missing files or invalid input do not crash the programme.

3.2 Test Evidence

```
$ python3 crypto_toolkit.py genkey
[OK] Key generated and saved to secret.key
...
```

4 Network Traffic Filtering

Firewall rules were applied in the authorised laboratory environment using `iptables` (see `firewall_rules.sh`).

4.1 Rules Summary

- Block guest subnet → records-server port
- Allow staff subnet → records-server port
- Drop all other inbound traffic to that port

4.2 Test Procedure and Results

Test	Source	Expected	Result
1	Staff subnet	Allowed	
2	Guest subnet	Blocked	
3	External/unauthorised	Blocked	

Table 2: Firewall test results

5 Conclusion

The measures implemented – network segmentation via firewall rules, authenticated symmetric encryption, and SHA-256 integrity verification – address the vulnerabilities identified in the risk assessment.

GitHub repository: <https://github.com/<your-username>/cryptography-network-security-exam>